

Python – Session 3 Practice Questions

Topics: If Conditions & Loops

Part 1 – If Conditions

1. Write a program that checks whether a number is positive, negative, or zero.
2. Write a program that checks whether a number is even or odd.
3. Ask the user for their age. Print whether they are eligible to vote (age 18 or above).
4. Write a program that asks for two numbers and prints the larger number.
5. Write a program that asks for three numbers and prints the largest one.
6. Ask the user for a student's grade and print: A for 90+, B for 80–89, C for 70–79, D for 60–69, and F below 60.
7. Write a program that checks whether a given year is a leap year.
8. Ask the user for a username and password. Print 'Login Successful' only if both are correct.
9. Write a program that checks whether a number is divisible by both 3 and 5.
10. Write a program that calculates a discount: if the price is greater than or equal to 1000, give 20% discount; otherwise give 10%.

Part 2 – For Loops

11. Print the numbers from 1 to 10 using a for loop.
12. Print all even numbers from 2 to 20 using a for loop.
13. Print all odd numbers from 1 to 19 using a for loop.
14. Ask the user for a number n and print numbers from 1 to n.
15. Calculate and print the sum of numbers from 1 to 100.
16. Ask the user for a number and print its multiplication table from 1 to 12.
17. Count how many numbers between 1 and 100 are divisible by 7.
18. Print the square of every number from 1 to 10.
19. Ask the user for 5 numbers and calculate their total sum.
20. Find the largest number in a list using a for loop without using max().

Part 3 – While Loops

21. Print numbers from 1 to 10 using a while loop.
22. Print numbers from 10 down to 1 using a while loop.
23. Ask the user for a positive number. Keep asking until they enter a positive number.

24. Create a program that keeps asking the user to enter a password until the correct password is entered.
25. Calculate the sum of numbers from 1 to n using a while loop.
26. Count the number of digits in an integer using a while loop.
27. Create a simple guessing game. The program stores a secret number, and the user keeps guessing until they find it.
28. Ask the user to enter numbers repeatedly. Stop when they enter 0, then print the sum of all entered numbers.

Part 4 – Nested Loops & Patterns

29. Use nested loops to print a 5×5 square of stars.
30. Print the following pattern using nested loops:

```
*  
**  
***  
****  
*****
```

31. Print a multiplication table from 1 to 5 using nested loops.
32. Print this pattern:
1
12
123
1234
12345

Part 5 – Mixed Challenges

33. Write a program that prints all prime numbers between 1 and 100.
34. Write a program that asks for a number and determines whether it is prime.
35. Ask the user for 10 numbers and count how many are positive, negative, and zero.
36. Create a program that calculates the factorial of a number using a loop.
37. Write a program that reverses a number using a while loop. Example: 1234 → 4321.
38. Create a simple ATM menu using a while loop: 1-Check Balance, 2-Deposit, 3-Withdraw, 4-Exit.
39. Write a program that asks for a student's name and 3 grades, calculates the average, and prints Pass if the average is at least 50; otherwise Fail.
40. Challenge: Create a number guessing game with a maximum of 5 attempts. Print whether the user won or lost.

Bonus – Debugging Questions

41. Find and fix the error: `for i in range(1, 10) print(i)`

42. Find and fix the error: `if age >= 18 print('Adult')`

43. What is wrong with this code?

```
while x < 10:
```

```
print(x)
```

Explain why it may become an infinite loop.

44. What will be the output?

```
for i in range(3):
```

```
print(i)
```

45. What will be the output?

```
for i in range(2, 10, 2):
```

```
print(i)
```

Instructor Note: Students should try to solve the questions independently first. Focus on understanding the condition, loop logic, `range()`, and when to use `for` vs. `while`.